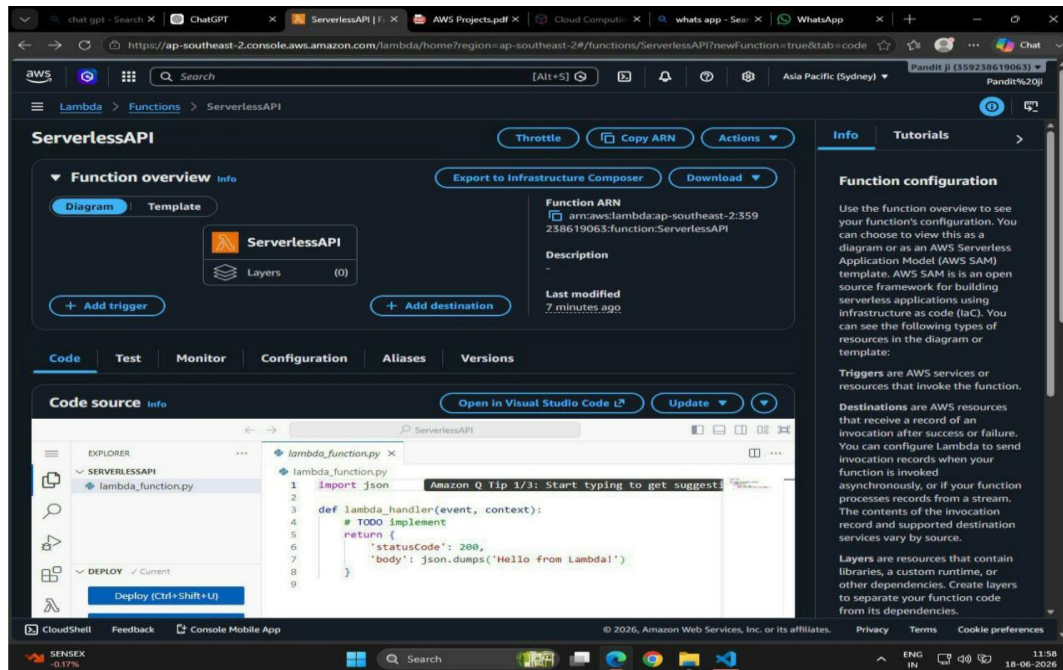


**Project 6: Serverless API with Lambda & API
Gateway**
**Build a RESTful API using AWS Lambda and API
Gateway**

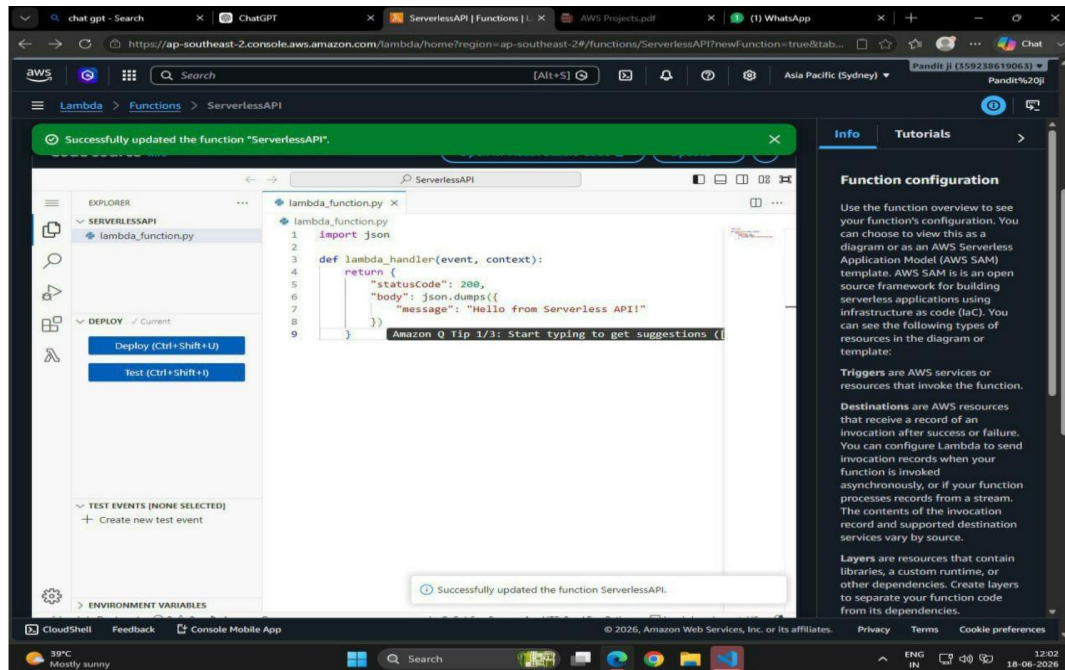
Project Screenshot 1



Step 1: Create an AWS Lambda Function

- Go to AWS Lambda Console.
- Click Create Function → Author from Scratch.
- Enter function name: ServerlessAPI.
- Choose Python 3.x or Node.js 18.x.
- Create a new role with basic Lambda permissions.
- Deploy sample code returning: Hello from Serverless API!

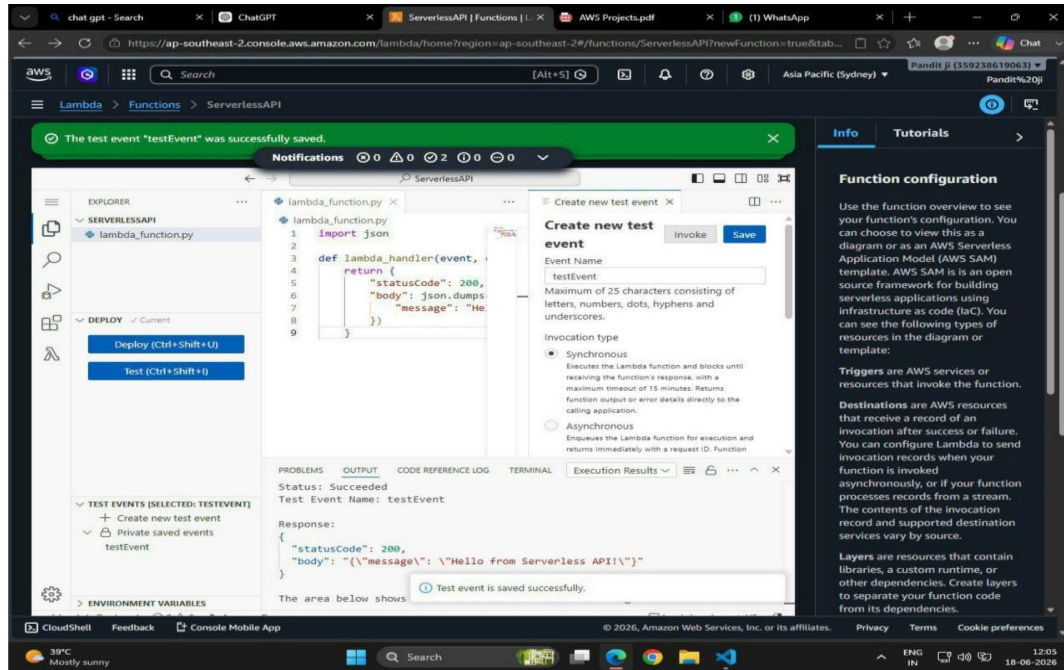
Project Screenshot 2



Step 2: Create an API in API Gateway

- Open API Gateway and Create API.
- Choose REST API and Build.
- Create a resource named hello.
- Create GET method.
- Integrate with the Lambda function.
- Save and confirm integration.

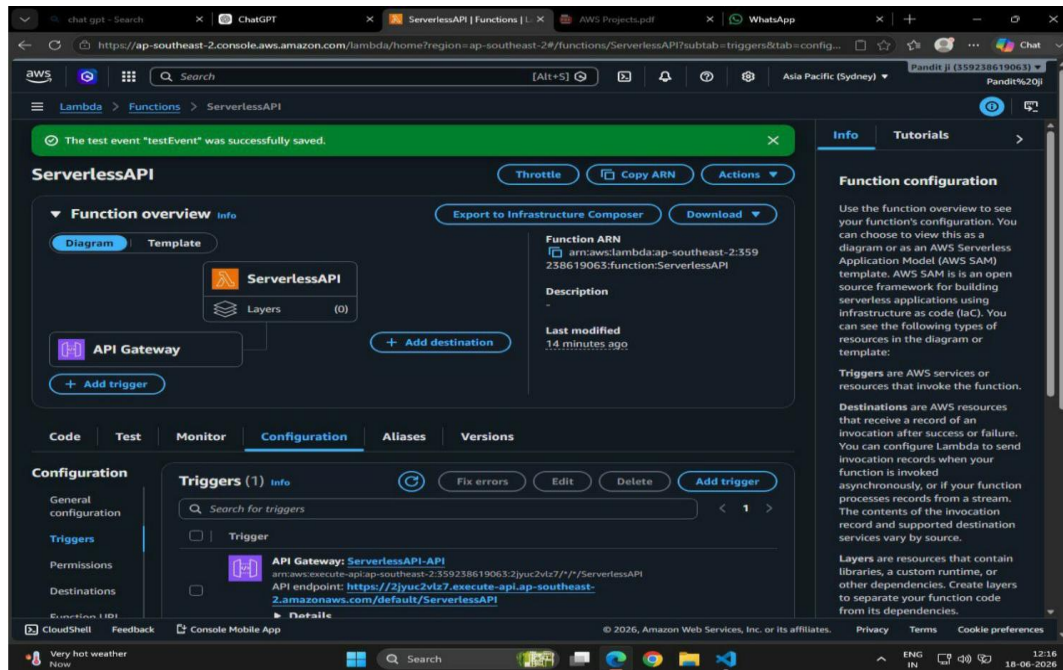
Project Screenshot 3



Step 3: Deploy the API

- Click Actions → Deploy API.
- Create stage: prod.
- Deploy the API.
- Copy the generated Invoke URL.

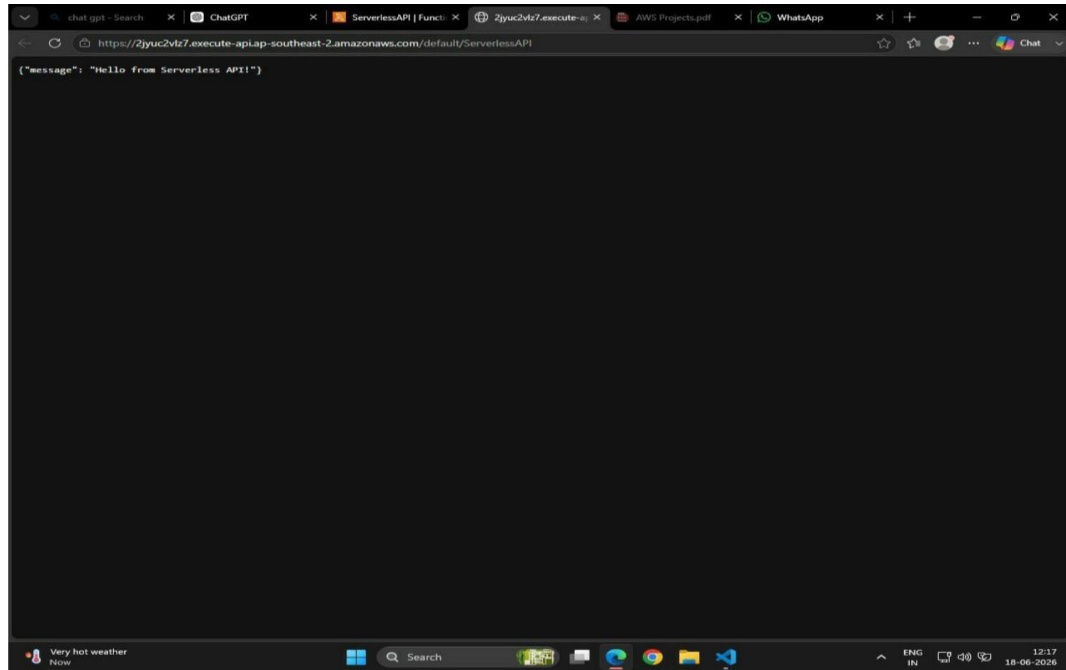
Project Screenshot 4



Step 4: Test the API

- Use Browser, Postman, or cURL.
- Send a GET request to the endpoint.
- Verify the JSON response.
- Expected Output: `{"message":"Hello from Serverless API!"}`

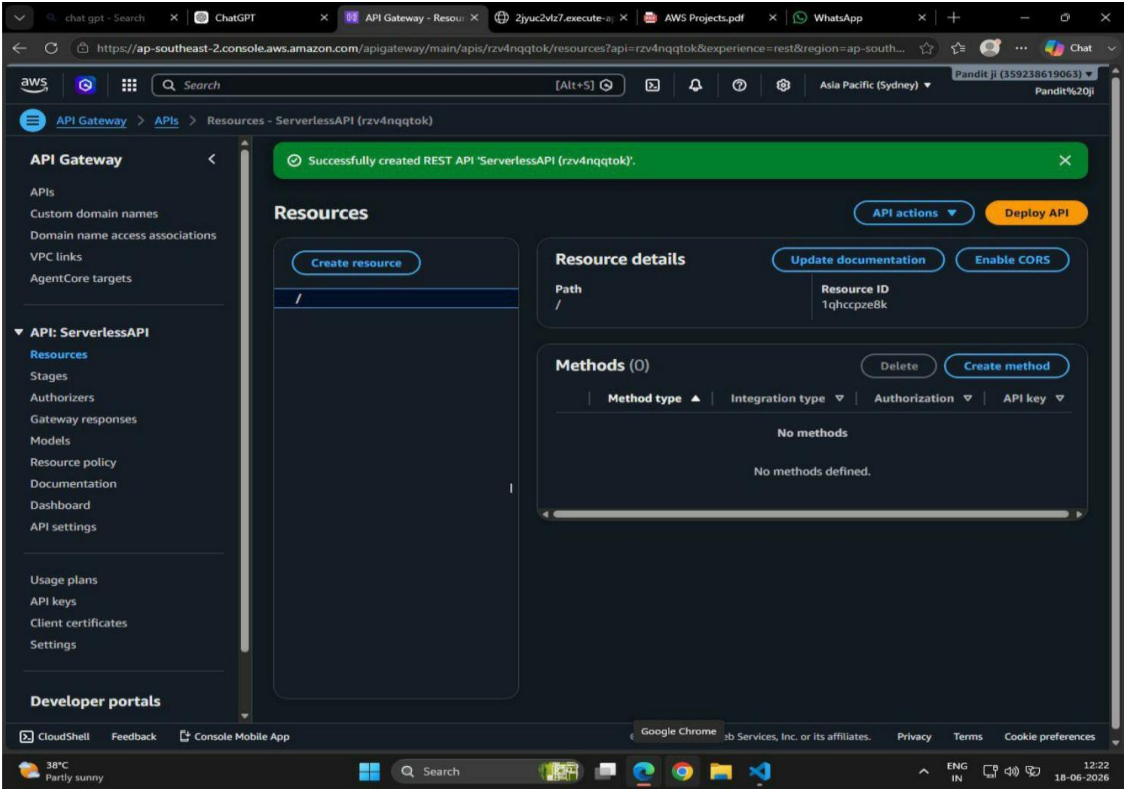
Project Screenshot 5



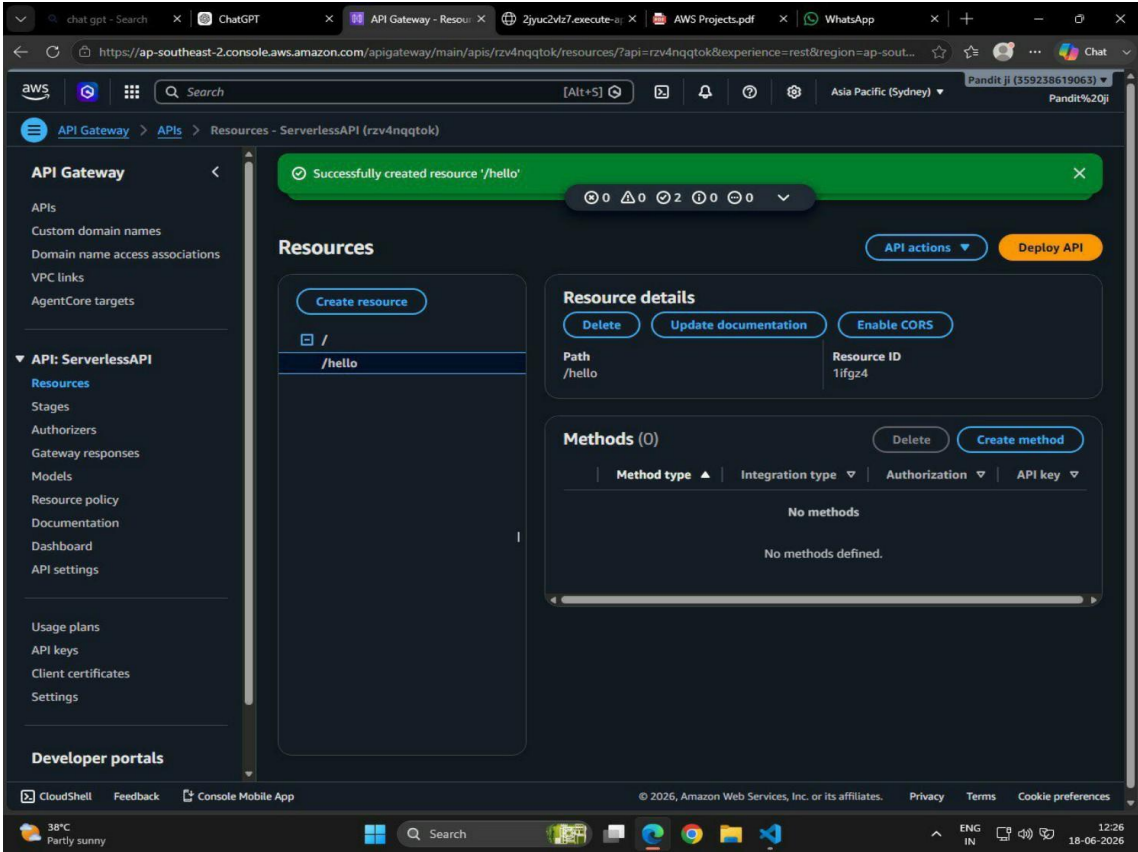
Enhancements (Optional)

- Integrate DynamoDB.
- Add JWT Authentication using Cognito.
- Enable CORS.
- Improve scalability and security.

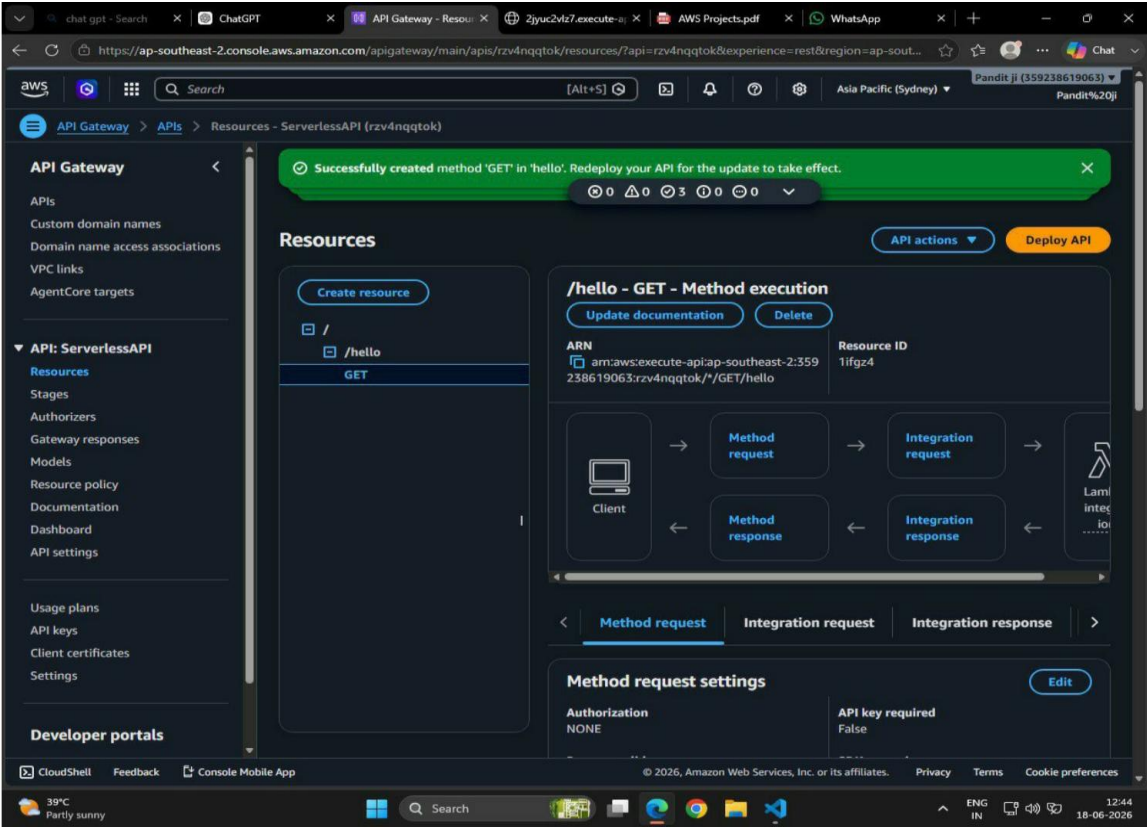
Project Screenshot 6



Project Screenshot 7



Project Screenshot 8



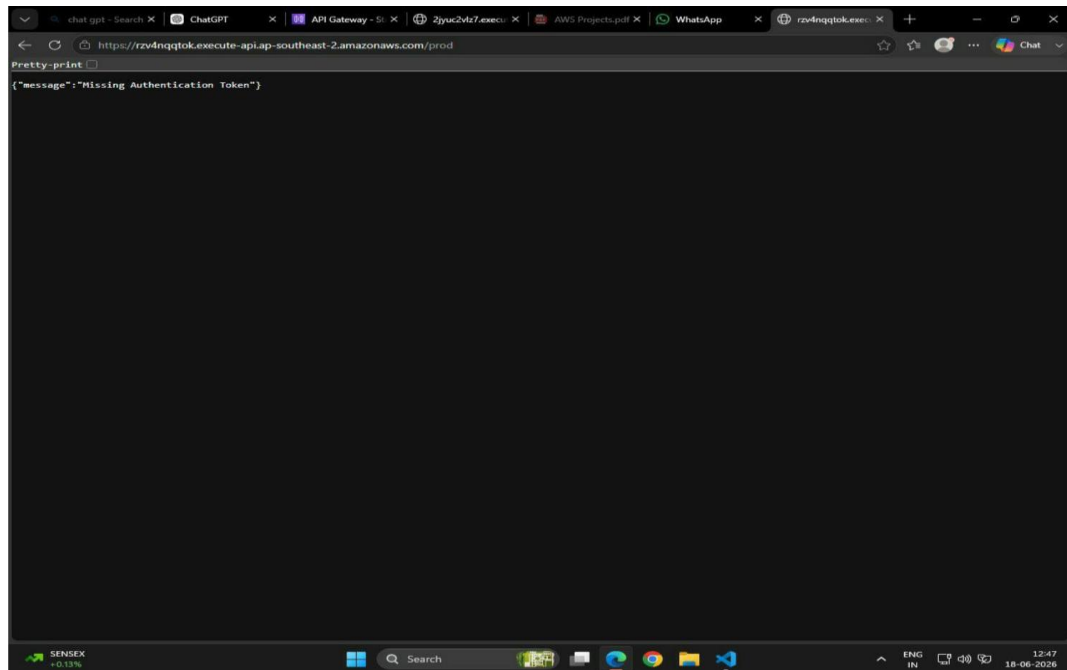
Project Screenshot 9

The screenshot displays the AWS API Gateway console for a ServerlessAPI. A green notification banner at the top states: "Successfully created deployment for ServerlessAPI. This deployment is active for prod." The left sidebar shows the navigation menu with "API Gateway" selected. The main content area is titled "Stages" and lists a single stage named "prod". The "Stage details" panel for the "prod" stage shows the following configuration:

- Stage name:** prod
- Web ACL:** -
- Burst info:** 5000
- Default method-level caching:** Inactive
- Rate info:** 10000
- Cache cluster info:** Inactive
- Client certificate:** -
- Invoke URL:** <https://rzv4nqqtok.execute-api.ap-southeast-2.amazonaws.com/prod>
- Active deployment:** uckwod on June 18, 2026, 12:45 (UTC+05:30)
- Logs and tracing:**
 - CloudWatch logs:** Inactive
 - Detailed metrics:** Inactive
 - Data tracing:** X-Ray tracing

The bottom of the console shows the footer with copyright information: "© 2026, Amazon Web Services, Inc. or its affiliates." and links for Privacy, Terms, and Cookie preferences. The system status bar at the very bottom indicates a temperature of 39°C and a "Partly sunny" weather condition.

Project Screenshot 10



Project Screenshot 11

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Student> curl https://rzv4nqgtok.execute-api.ap-southeast-2.amazonaws.com/prod/hello

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is
parsed.
RECOMMENDED ACTION:
Use the -UseBasicParsing switch to avoid script code execution.
Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): A

StatusCode      : 200
StatusDescription : OK
Content          : {"statusCode": 200, "body": "{\"message\": \"Hello from Serverless API!\"}"}
RawContent       : HTTP/1.1 200 OK
                  Connection: keep-alive
                  x-amzn-RequestId: 807386e5-c56e-4ce3-92cb-917fc63e77f1
                  x-amz-apigw-id: fJZniEb8SwMEopA=
                  X-Amzn-Trace-Id: Root=1-6a33a2fc-24d83c0145ec2alc3e0346ec;Parent=7729...
Forms            : {}
Headers          : [{"Connection", "keep-alive"}, [{"x-amzn-RequestId", "807386e5-c56e-4ce3-92cb-917fc63e77f1"},
                  [{"x-amz-apigw-id", "fJZniEb8SwMEopA="}, [{"X-Amzn-Trace-Id", "Root=1-6a33a2fc-24d83c0145ec2alc3e0346ec;Pare
nt=77290c1ead6de9c9;Sampled=0;Lineage=1:ab7ed6fc:0}]]]}
Images           : {}
InputFields      : {}
Links            : {}
ParsedHtml       : mshtml.HTMLDocumentClass
RawContentLength : 76

PS C:\Users\Student> |
```

PROJECT COMPLETION

**This Serverless API Project was successfully
documented and compiled by:**

Raunak srivastava

Devansh Mishra

Priyanshu roy

Ankit singh

**With dedication, teamwork, innovation, and a
passion for cloud computing and AWS
technologies.**

***"Transforming ideas into scalable cloud
solutions."***